



Lower bounds for treewidth of product graphs



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ABSTRACT

Two lower bounds for the treewidth of product graphs are presented in terms of the bramble number. The first bound is that the bramble number of the Cartesian product of graphs G_1 and G_2 must be at least the product of the Hadwiger number of G_1 and the PI number of G_2 , where the PI number is a new graph parameter introduced in this paper. The second bound is that the bramble number of the strong product of graphs G_1 and G_2 must be at least the product of the Hadwiger number of G_1 and the bramble number of G_2 . We also demonstrate applications of the lower bounds.

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1. Introduction

The concept of *treewidth* has contributed greatly to pure and algorithmic graph theories in the recent decades. In rough terms, the treewidth of a graph G , denoted by $\text{tw}(G)$, is a graph parameter that measures the proximity of G to a tree. In this paper, we present two lower bounds for the treewidth of product graphs. For this purpose, instead of the treewidth, we use another graph parameter known as the bramble number, which is essentially the same as the treewidth. A *bramble* $\mathcal{B} = \{B_1, \dots, B_{|\mathcal{B}|}\}$ of G is a collection of the vertex sets of connected subgraphs of G such that any B_i and B_j in $\mathcal{B}$ intersect or are *joined* by an edge. The *order* of $\mathcal{B}$ is the least number of vertices required to cover every B_i in $\mathcal{B}$. In other words, it is the size of a minimum hitting set of $\mathcal{B}$. The *bramble number* of a graph G , denoted by $\text{bn}(G)$, is the maximum order of all brambles of G . Seymour and Thomas [22] showed that $\text{bn}(G) = \text{tw}(G) + 1$ for any graph G . A merit of using a bramble is that a lower bound for the treewidth may be found *constructively*. That is, if we construct a bramble of order greater than k , then the bramble is a certificate of a lower bound k of the treewidth.

1.1. Motivation

Our study was motivated by the following natural question that arises from a study of the inapproximability of the bramble number: is there a graph product operation under which the treewidth of a resulting product graph can be determined only by the treewidths of its factor graphs? We explain how this question relates to the inapproximability by providing a famous example. The *clique number* $\omega(G)$ of a graph G is the size of a maximum clique in G . The *strong product* of $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, denoted by $G_1 \boxtimes G_2$, is the graph with the vertex set $V_1 \times V_2$, in which two vertices (u_1, v_1) and (u_2, v_2) are adjacent if and only if $u_1 = u_2$ or $\{u_1, u_2\} \in E_1$, and $v_1 = v_2$ or $\{v_1, v_2\} \in E_2$ (see Fig. 1 for an example). It is

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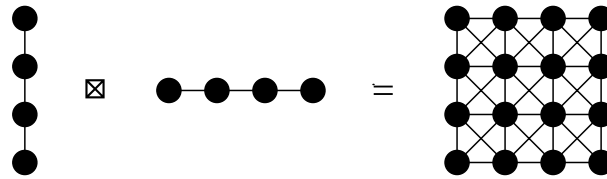


Fig. 1. The strong product of two paths.

not difficult to observe that $\omega(G_1 \boxtimes G_2) = \omega(G_1) \cdot \omega(G_2)$ for any graphs G_1 and G_2 . Hence, if we denote the strong power of k copies of G as G' , then $\omega(G') = (\omega(G))^k$. Furthermore, it is known that for a given clique C' of G' , we can compute a clique C of G with size at least $|C'|^{1/k}$ in polynomial time. This fact can be used to amplify the approximation hardness of the clique number as follows (see [1, Section 6.4]). Assume that there is an α -approximation algorithm $\mathcal{A}$ for the clique number with a constant approximation ratio $\alpha > 1$; that is, for any input, $\mathcal{A}$ outputs a clique of size at least $1/\alpha$ of the maximum. Let $\mathcal{A}'$ be the following (polynomial-time) algorithm:

1. compute the strong power G' of k copies of G ;
2. apply $\mathcal{A}$ to G' and obtain a clique C' of size at least $\alpha^{-1} \cdot \omega(G')$;
3. compute a clique C of G with size at least $|C'|^{1/k}$ from C' ;
4. output C .

From the above discussion, it follows that

$$|C| \geq |C'|^{1/k} \geq (\alpha^{-1} \cdot \omega(G'))^{1/k} = (\alpha^{-1} \cdot \omega(G)^k)^{1/k} = \alpha^{-1/k} \cdot \omega(G).$$

This implies that $\mathcal{A}'$ is an $\alpha^{1/k}$ -approximation algorithm for the clique number. Therefore, for any $r > 1$, we can obtain a polynomial-time r -approximation algorithm for the clique number by setting $k \geq \log \alpha / \log r$. That is, $\mathcal{A}'$ is a PTAS for the maximum clique problem. On the other hand, it can be shown using the PCP theorem that such a PTAS does not exist unless $P = NP$ (see [1, Section 6.4]). Hence, there is no constant-factor approximation algorithm for the clique number, unless $P = NP$. Note that the PCP theorem allows us to have a stronger approximation hardness for clique. See Håstad's result [15] that shows the $n^{1-\epsilon}$ approximation hardness for any $\epsilon > 0$.

The approximability of the treewidth (and thus, that of the bramble number) is well studied. The best known approximation ratio $O(\sqrt{\log \text{opt}})$, where opt is the optimum value, was derived by Feige, Hajiaghayi, and Lee [12]. On the other hand, inapproximability is a long-standing open issue. The only known fact is the hardness of additive error approximation. Bodlaender, Gilbert, Hafsteinsson, and Kloks [4] showed that no polynomial-time algorithm $\mathcal{A}$ for the treewidth of a graph G can guarantee $\mathcal{A}(G) \leq \text{tw}(G) + |V(G)|^\epsilon$ for any constant $\epsilon < 1$ unless $P = NP$. It is not known whether the problem admits a PTAS or a constant factor approximation algorithm. If we have a graph product $\otimes$ such that $\text{tw}(G_1 \otimes G_2) = \text{tw}(G_1) \cdot \text{tw}(G_2)$ or at least $\text{tw}(G \otimes G) = (\text{tw}(G))^2$, and we also have a polynomial-time algorithm for computing a tree-decomposition of G with width at most $w^{1/k}$ from a tree-decomposition of $\bigotimes_{1 \leq i \leq k} G$ with width w , then having a constant factor approximation is equivalent to having a PTAS for treewidth. This may help in the study of inapproximability of treewidth.

Quite recently, Austrin, Pitassi, and Wu [2] have proved, assuming the recently introduced Small Set Expansion (SSE) conjecture [19], that approximating the treewidth of a graph in any constant factor is NP-hard. They mentioned that since the status of the SSE conjecture is very uncertain, their inapproximability result should not be taken as the absolute evidence that there is no constant factor approximation for treewidth.

1.2. Our results

Our original motivation was to study the possibility of the existence of a graph product operation $\otimes$ such that $\text{bn}(G_1 \otimes G_2) = \text{bn}(G_1) \cdot \text{bn}(G_2)$ for any graphs G_1 and G_2 , or at least, $\text{bn}(G \otimes G) = (\text{bn}(G))^2$ for any graph G . This work presents the first step for this direction in the study of the inapproximability of the bramble number (and treewidth). In this work, we present two lower bounds for the bramble number of product graphs in terms of two related graph parameters. We first show that the bramble number of the Cartesian product of graphs G_1 and G_2 is at least the product of the *PI number* of G_1 and the *Hadwiger number* of G_2 ; that is, in our terminology, $\text{bn}(G_1 \square G_2) \geq \iota(G_1) \cdot \eta(G_2)$, where $\iota(G)$ and $\eta(G)$ denote the PI number and the Hadwiger number of G , respectively. Next, we show, using a similar argument, that the bramble number of the strong product of graphs G_1 and G_2 is at least the product of the bramble number of G_1 and the Hadwiger number of G_2 ; that is, $\text{bn}(G_1 \boxtimes G_2) \geq \text{bn}(G_1) \cdot \eta(G_2)$. See Section 2 for the definitions and descriptions of notations used.

We also demonstrate applications of the lower bounds. By applying one lower bound, we determine, with an additive error of 1, the treewidth of the Cartesian product graph of a complete graph and a grid. We also apply the lower bound to the Cartesian product graph of a complete graph and a complete multipartite graph. Unfortunately, our two lower bounds are very weak in the case where both the factor graphs have small treewidth. For example, although $\text{bn}(P_n \square P_n) = n + 1$ and $\text{bn}(P_n \boxtimes P_n) \geq n + 1$, the lower bound functions give $\text{bn}(P_n \square P_n) \geq \iota(P_n) \cdot \eta(P_n) = 2$ and $\text{bn}(P_n \boxtimes P_n) \geq \text{bn}(P_n) \cdot \eta(P_n) = 4$, where P_n is the path of n vertices. On the other hand, these lower bounds work well in the case where one of the two factor

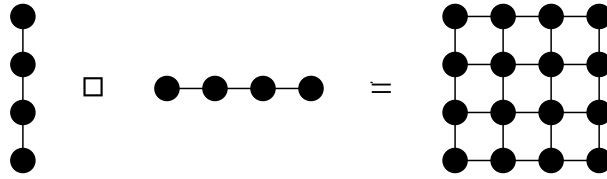


Fig. 2. The Cartesian product of two paths.

graphs has large treewidth, such as that in the case of a complete graph. In particular, the lower bound for the strong product precisely determines the bramble number if one of the operands is a complete graph.

1.3. Related work

Djelloul [9,10] presented upper bounds for the treewidth of Cartesian product graphs. Bodlaender, Gilbert, Hafsteinsson, and Kloks [4] showed implicitly that $\text{bn}(G \boxtimes K_n) = \text{bn}(G) \cdot n$ to demonstrate inapproximability with an absolute error guarantee for the treewidth. Our lower bound $\text{bn}(G_1 \boxtimes G_2) \geq \text{bn}(G_1) \cdot \eta(G_2)$ can be seen as a generalization of the fact $\text{bn}(G \boxtimes K_n) = \text{bn}(G) \cdot \eta(K_n)$. Lucena [18] determined the treewidth of the Cartesian product of two copies of K_n . More precisely, it was shown that $\text{bn}(K_n \square K_n) = (n^2 + n)/2$ for $n \geq 3$, where $G_1 \square G_2$ denotes the Cartesian product of two graphs G_1 and G_2 . Our lower bound can be considered as a generalization of the lower bound on $\text{bn}(K_n \square K_n)$ for odd n (see Section 4). Recently, Wood [24] has presented a lower bound for the treewidth of Cartesian products of highly connected graphs. In particular, he showed that for all k -connected graphs G and H , each with at least n vertices, the treewidth of the Cartesian product of G and H is at least $k(n - 2k + 2) - 1$. This result generalizes the known fact that $n \times n$ grids have treewidth n , up to an additive error of -1 .

1.4. Organization of this paper

The rest of the paper is organized as follows. Section 2 reviews basic definitions and notations and introduces new terms. Section 3 presents the lower bounds for the treewidth of product graphs. Section 4 demonstrates the applications of the lower bounds. In Section 5, we make a simple observation on a relationship between the Hadwiger number and the bramble number.

2. Definitions and notations

In this work, we only consider undirected simple finite graphs. Let G be a graph. We denote as $V(G)$ and $E(G)$, the vertex set and the edge set of G , respectively. Let X and Y be subsets of $V(G)$. The induced subgraph of G induced by X is denoted by $G[X]$. If $G[X]$ is connected, then we say that X is a *connected subset* or simply *connected*. Two sets X and Y are *joined* if X and Y do not intersect and there exists an edge that has one endpoint in X and another in Y . Two sets X and Y are *touched* if they intersect or are joined. Note that according to our definitions, X and Y are joined only if they do not intersect.

For two sets A and B , let $A \times B$ denote the set $\{(a, b) \mid a \in A, b \in B\}$. Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be graphs. The *Cartesian product* $G_1 \square G_2$ is the graph whose vertex set is $V_1 \times V_2$, and any two vertices (u_1, v_1) and (u_2, v_2) are adjacent in $G_1 \square G_2$ if and only if either $v_1 = v_2$ and $\{u_1, u_2\} \in E_1$, or $u_1 = u_2$ and $\{v_1, v_2\} \in E_2$ (see Fig. 2). It is evident that $G_1 \square G_2$ is a spanning subgraph of $G_1 \boxtimes G_2$.

We denote as P_n and K_n , the path of n vertices and the complete graph of n vertices, respectively. Let $K_{n_1, \dots, n_k}$ denote the complete multipartite graph with k parts, where n_i is the number of vertices in the i th part of the graph.

2.1. Graph parameters

A *tree-decomposition* of a graph G is a pair $(\mathcal{X}, T)$, where $\mathcal{X} = \{X_i \mid i \in V(T)\}$ is a collection of subsets of $V(G)$ and T is a tree such that

- $\bigcup_{i \in V(T)} X_i = V(G)$;
- for each edge $\{v, w\} \in E(G)$, there is a node $i \in V(T)$ such that $v, w \in X_i$; and
- for each $v \in V(G)$, the set of nodes $\{i \mid v \in X_i\}$ forms a subtree of T .

The elements in $\mathcal{X}$ are called *bags*. The *width* of a tree-decomposition $(\mathcal{X}, T)$ is $\max_{i \in V(T)} |X_i| - 1$. The *treewidth* of G , denoted by $\text{tw}(G)$, is the minimum width over all tree-decompositions of G .

Let $\mathcal{S} = \{S_1, \dots, S_k\}$ be a collection of connected subsets of $V(G)$. The *order* of $\mathcal{S}$ is the size of a minimum *hitting set* of $\mathcal{S}$; that is, the minimum size of a subset $H \subseteq V(G)$ that intersects every subset $S_i \in \mathcal{S}$. The family $\mathcal{S}$ is an *intersecting family*, a *joined family*, or a *bramble* if any two subsets in $\mathcal{S}$ intersect, are joined, or are touched, respectively (see Fig. 3). Recall that all subsets in $\mathcal{S}$ require to be connected. It is obvious that given a bramble, determining its order is NP-hard: any family of

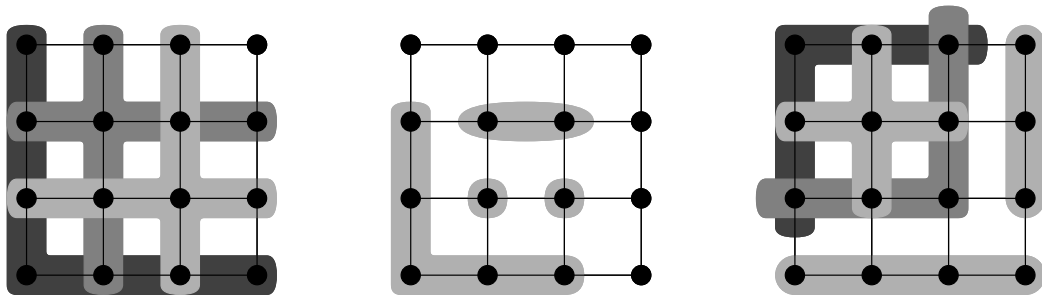


Fig. 3. An intersecting family, a joined family, and a bramble of a grid.

vertex subsets of a complete graph is a bramble; and the problem of determining the size of a minimum hitting set of a given family is NP-hard [13]. Furthermore, even if the given family is an intersecting family, the problem of determining its order remains NP-hard [14]. In general, it is difficult to obtain a good lower bound for the order of $\mathcal{F}$ when $\mathcal{F}$ is an intersecting family or a bramble [17]. On the other hand, we can easily determine the order of a joined family, given that its order is simply its cardinality.

The following graph parameters play central roles in this work, and there are several known results on the Hadwiger number of product graphs (see e.g. [7,23]) and the bramble number (see [5]).

Definition 2.1. The *PI number* of G , denoted as $\iota(G)$, is the maximum order over all intersecting families of G . (PI stands for “Pairwise Intersecting”.) The *Hadwiger number* of G , denoted as $\eta(G)$, is the maximum order over all joined families of G . The *bramble number* of G , denoted as $\text{bn}(G)$, is the maximum order over all brambles of G .

From the definitions above, it is evident that $\text{bn}(G) \geq \max\{\iota(G), \eta(G)\}$. The Hadwiger number of a graph G can be defined as the maximum number k such that G has K_k as a minor. This implies that the Hadwiger number is at most 4 for any planar graph. On the other hand, it can be shown that the PI number is unbounded for planar graphs (see the proof of Theorem 4.1). It is known that computing the Hadwiger number of a given graph is NP-hard [11]. To the best of our knowledge, the complexity for determining the PI number of a graph is not known.

The famous conjecture “ $\eta(G) \geq \chi(G)$ for any graph G ” is known as Hadwiger’s conjecture, where $\chi(G)$ denotes the chromatic number of G (see [6]). Since $\text{bn}(G) \geq \eta(G)$, it is natural to ask whether $\text{bn}(G) \geq \chi(G)$ holds for any graph G . Actually, this weaker inequality holds for any graph. To see this, consider a chordal graph G' such that G' is a supergraph of G and $\text{bn}(G') = \text{bn}(G)$. Then, since G' is a chordal graph, $\chi(G') = \omega(G')$ holds, where $\omega(G')$ denotes the clique number of G' . Thus we have $\text{bn}(G) = \text{bn}(G') = \omega(G') = \chi(G') \geq \chi(G)$. Consequently, Hadwiger’s conjecture holds for any graph G for which $\text{bn}(G) = \eta(G)$.

As mentioned in the introduction, Seymour and Thomas [22] showed that

$$\text{bn}(G) = \text{tw}(G) + 1$$

for any graph G . There is an advantage of using a bramble for presenting a lower bound on the treewidth. This is because, by considering a suitable bramble, the evaluation of whose order is easy, we can constructively obtain the certification of a lower bound for the treewidth. For example, Bodlaender, Grigoriev, and Koster [5] presented a very simple proof of the lower bound for the treewidth of grids by using brambles.

2.2. Remarks on PI number

It should be mentioned that the graph parameter *linkedness* is strongly related to the PI number. In this work, we will use an idea inherent in the concept of the graph parameter linkedness in order to obtain a lower bound for $\iota(G)$. The definition of linkedness is as follows.

Definition 2.2 ([20,21]). A subset S of $V(G)$ is *k-linked* if for any set $X \subseteq V(G)$ with $|X| < k$, there is a component of $G - X$ containing more than half of the vertices of S . The *linkedness* of G , denoted as $\lambda(G)$, is the largest k for which G has a *k-linked* set.

A prime example of linkedness is $\lambda(K_n) = \lceil n/2 \rceil$. This is because $V(K_n)$ is an $\lceil n/2 \rceil$ -linked set.

For each set $X \subseteq V(G)$ with $|X| < k$ and a *k-linked* set S , we call the component of $G - X$ containing more than half of the vertices of S the *big component* of $G - X$ under S .

Remark 2.3. The set of big components under S intersect pairwise because each of them has more than half of the vertices of S .

Remark 2.4. The order of the set of big components under a *k-linked* set S is at least k . This is because otherwise, there is a hitting set X of size less than k . From the definition, $G - X$ should have a big component and clearly, X and the big component do not intersect.

Because each big component is connected and they intersect pairwise, as mentioned in Remark 2.3, the set of big components is an intersecting family. Thus, from Remark 2.4, we have $\lambda(G) \leq \iota(G)$. In Section 4, we present a lower bound for the PI number of a complete multipartite graph by using the idea introduced in Remark 2.4.

It is known that for any graph G , $\text{bn}(G) \leq 2\lambda(G)$ [20], and hence, $\iota(G) \leq \text{bn}(G) \leq 2\iota(G)$. Thus the PI number of a graph approximates its bramble number within a factor of 2.

3. Lower bounds

In this section, we present the main results of this work. That is, we shall prove the following two theorems.

Theorem 3.1. For any graphs G_1 and G_2 , $\text{bn}(G_1 \square G_2) \geq \max\{\iota(G_1) \cdot \eta(G_2), \iota(G_2) \cdot \eta(G_1)\}$.

Theorem 3.2. For any graphs G_1 and G_2 , $\text{bn}(G_1 \boxtimes G_2) \geq \max\{\text{bn}(G_1) \cdot \eta(G_2), \text{bn}(G_2) \cdot \eta(G_1)\}$.

Theorem 3.1 and the fact $\text{bn}(G) \leq 2\iota(G)$ result in the following corollary.

Corollary 3.3. For any graphs G_1 and G_2 , $\text{bn}(G_1 \square G_2) \geq \frac{1}{2} \cdot \max\{\text{bn}(G_1) \cdot \eta(G_2), \text{bn}(G_2) \cdot \eta(G_1)\}$.

To prove the two main theorems, we need the following three lemmas.

Lemma 3.4. If $\mathcal{I} = \{I_1, \dots, I_{|\mathcal{I}|}\}$ is an intersecting family of a graph G_1 and $\mathcal{F} = \{F_1, \dots, F_{|\mathcal{F}|}\}$ is a bramble of a graph G_2 , then $\mathcal{B} = \{I_i \times F_j \mid I_i \in \mathcal{I}, F_j \in \mathcal{F}\}$ is a bramble of $G_1 \square G_2$.

Proof. It is known that the Cartesian product of two graphs is connected if and only if both the graphs are connected (see [16]). Because both $G_1[I_i]$ and $G_2[F_j]$ are connected, it follows that $(G_1 \square G_2)[I_i \times F_j] = (G_1[I_i]) \square (G_2[F_j])$ is connected. Thus each $I_i \times F_j$ is connected.

Now, let us check whether any $I_i \times F_j$ and $I_p \times F_q$ are touched. Given that $\mathcal{I}$ is an intersecting family, there exists a vertex $u \in I_i \cap I_p$. If there is an element $v \in F_j \cap F_q$, then $(u, v) \in (I_i \times F_j) \cap (I_p \times F_q)$, and thus, $I_i \times F_j$ and $I_p \times F_q$ intersect. If $F_j \cap F_q = \emptyset$, then there is an edge $\{v, w\}$ in G_2 such that $v \in F_j$ and $w \in F_q$, and hence, $(u, v) \in B_{i,j}$ and $(u, w) \in B_{p,q}$ are adjacent in $G_1 \square G_2$. $\square$

Lemma 3.5. If $\mathcal{S} = \{S_1, \dots, S_{|\mathcal{S}|}\}$ is a bramble of a graph G_1 and $\mathcal{T} = \{T_1, \dots, T_{|\mathcal{T}|}\}$ is a bramble of a graph G_2 , then $\mathcal{B} = \{S_i \times T_j \mid S_i \in \mathcal{S}, T_j \in \mathcal{T}\}$ is a bramble of $G_1 \boxtimes G_2$.

Proof. It is easy to see that $(G_1 \square G_2)[S_i \times T_j]$ is a spanning subgraph of $(G_1 \boxtimes G_2)[S_i \times T_j]$. Therefore, $(G_1 \boxtimes G_2)[S_i \times T_j]$ is connected because $(G_1 \square G_2)[S_i \times T_j]$ is connected (see the first half of the proof of Lemma 3.4). Thus $S_i \times T_j$ is connected.

Let $S_i \times T_j$ and $S_p \times T_q$ be elements of $\mathcal{B}$. If $S_i \cap S_p \neq \emptyset$ or $T_j \cap T_q \neq \emptyset$, then we can show that $S_i \times T_j$ and $S_p \times T_q$ are touched by using the same argument as that used in the second half of the proof of Lemma 3.4. Thus we assume $S_i \cap S_p = \emptyset$ and $T_j \cap T_q = \emptyset$. Now there exist edges $\{u, v\} \in E(G_1)$ and $\{x, y\} \in E(G_2)$ such that $u \in S_i$, $v \in S_p$, $x \in T_j$, and $y \in T_q$. From the definition of strong product, $G_1 \boxtimes G_2$ has the edge $\{(u, x), (v, y)\}$. Therefore, $S_i \times T_j$ and $S_p \times T_q$ are joined. $\square$

Lemma 3.6. Let $\mathcal{F} = \{F_1, \dots, F_{|\mathcal{F}|}\}$ be a family of sets and $\mathcal{J} = \{J_1, \dots, J_{|\mathcal{J}|}\}$ be a family of pairwise disjoint sets. If $\mathcal{B} = \{F_i \times J_j \mid F_i \in \mathcal{F}, J_j \in \mathcal{J}\}$, then

$$\text{the order of } \mathcal{B} = \text{the order of } \mathcal{F} \cdot |\mathcal{J}|.$$

Proof. Since $J_{j_1} \cap J_{j_2} = \emptyset$, it holds for $j_1 \neq j_2$ that

$$\left(\bigcup_{1 \leq i \leq |\mathcal{F}|} F_i \times J_{j_1} \right) \cap \left(\bigcup_{1 \leq i \leq |\mathcal{F}|} F_i \times J_{j_2} \right) = \emptyset.$$

Thus if we partition $\mathcal{B}$ into $\mathcal{B}_1, \dots, \mathcal{B}_{|\mathcal{J}|}$, where $\mathcal{B}_j = \{F_i \times J_j \mid 1 \leq i \leq |\mathcal{F}|\}$, then $(H \cap \mathcal{B}_{j_1}) \cap (H \cap \mathcal{B}_{j_2}) = \emptyset$ for any hitting set H of $\mathcal{B}$ when $j_1 \neq j_2$. Therefore,

$$\text{the order of } \mathcal{B} = \sum_{1 \leq j \leq |\mathcal{J}|} \text{the order of } \mathcal{B}_j.$$

It suffices to show that the order of $\mathcal{F}$ is equal to the order of $\mathcal{B}_j$ for each j . Let $H_{\mathcal{F}}$ be a hitting set of $\mathcal{F}$ and v be any element of J_j . Obviously, $H_{\mathcal{F}} \times \{v\}$ is a hitting set of $\mathcal{B}_j$. Hence the order of $\mathcal{B}_j$ is at most the order of $\mathcal{F}$. Conversely, let $H_{\mathcal{B}_j} = \{(u_1, v_1), \dots, (u_k, v_k)\}$ be a hitting set of $\mathcal{B}_j$, and let $F_p \in \mathcal{F}$. Given that $F_p \times J_j \in \mathcal{B}_j$, there exists an index r such that $(u_r, v_r) \in F_p \times J_j$. This implies that $u_r \in F_p$. Hence, the underlying set of $\{u_1, \dots, u_k\}$ is a hitting set of $\mathcal{F}$ with at most k elements. (Note that $\{u_1, \dots, u_k\}$ may be a multiset.) Therefore, the order of $\mathcal{F}$ is at most the order of $\mathcal{B}_j$. $\square$

Now, we are ready to prove the main theorems.

Proof of Theorem 3.1. Let $\mathcal{I} = \{I_1, \dots, I_{|\mathcal{I}|}\}$ be an intersecting family of G_1 with order $\iota(G_1)$, $\mathcal{J} = \{J_1, \dots, J_{\eta(G_2)}\}$ be a joined family of G_2 , and $\mathcal{B} = \{I_i \times J_j \mid I_i \in \mathcal{I}, J_j \in \mathcal{J}\}$. By Lemma 3.4, $\mathcal{B}$ is a bramble of $G_1 \square G_2$. By Lemma 3.6, the order of $\mathcal{B}$ is $\iota(G_1) \cdot \eta(G_2)$, given that the order of $\mathcal{I}$ is $\iota(G_1)$ and that $|\mathcal{J}| = \eta(G_2)$. Therefore, $\text{bn}(G_1 \square G_2) \geq \iota(G_1) \cdot \eta(G_2)$. Since the Cartesian product operation is commutative, it also holds that $\text{bn}(G_1 \square G_2) \geq \eta(G_1) \cdot \iota(G_2)$. $\square$

Proof of Theorem 3.2. Let $\mathcal{F} = \{F_1, \dots, F_{|\mathcal{F}|}\}$ be a bramble of G_1 with order $\text{bn}(G_1)$, let $\mathcal{J} = \{J_1, \dots, J_{\eta(G_2)}\}$ be a joined family of G_2 , and let $\mathcal{B} = \{F_i \times J_j \mid F_i \in \mathcal{F}, J_j \in \mathcal{J}\}$. By Lemma 3.5, $\mathcal{B}$ is a bramble of $G_1 \boxtimes G_2$. By Lemma 3.6, the order of $\mathcal{B}$ is $\text{bn}(G_1) \cdot \eta(G_2)$, given that the order of $\mathcal{F}$ is $\text{bn}(G_1)$ and that $|\mathcal{J}| = \eta(G_2)$. Therefore, $\text{bn}(G_1 \boxtimes G_2) \geq \text{bn}(G_1) \cdot \eta(G_2)$. Since the strong product operation is commutative, it also holds that $\text{bn}(G_1 \square G_2) \geq \eta(G_1) \cdot \text{bn}(G_2)$. $\square$

We used intersecting families and joined families in the proofs above. The following proposition demonstrates that they are necessary for our purpose.

Proposition 3.7. Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be families of subsets, and let $\mathcal{F}_{12} = \{S_1 \times S_2 \mid S_1 \in \mathcal{F}_1, S_2 \in \mathcal{F}_2\}$. (i) If $\mathcal{F}_1$ and $\mathcal{F}_2$ are intersecting families, then the order of $\mathcal{F}_{12}$ may be less than the order of $\mathcal{F}_1$ times the order of $\mathcal{F}_2$. (ii) If both $\mathcal{F}_1$ and $\mathcal{F}_2$ are joined families of graph G_1 and G_2 , respectively, then $\mathcal{F}_{12}$ may not be a bramble of $G_1 \square G_2$.

Proof. Let G_1 and G_2 be complete graphs of $2k + 1$ vertices for some $k \geq 1$, and let $\{u_1, \dots, u_{2k+1}\}$ and $\{v_1, \dots, v_{2k+1}\}$ be the vertex sets of G_1 and G_2 , respectively.

- (i) Let $\mathcal{I}_i$ be the family $\{S \subset V(G_i) \mid |S| = k + 1\}$ for $i \in \{1, 2\}$, and $\mathcal{I}_{12}$ be the set $\{S_1 \times S_2 \mid S_1 \in \mathcal{I}_1, S_2 \in \mathcal{I}_2\}$. It is evident that both $\mathcal{I}_1$ and $\mathcal{I}_2$ are intersecting families. From Remark 2.4, the order of $\mathcal{I}_i$ is at least $k + 1$. Thus the order of $\mathcal{I}_1$ times the order of $\mathcal{I}_2$ is at least $(k + 1)^2$. However, $\{(u_i, v_i) \mid 1 \leq i \leq 2k + 1\}$ is a hitting set of $\mathcal{I}_{12}$. This is because, for any element $S_1 \times S_2$ in $\mathcal{I}_{12}$, there is an index $i \in \{j \mid u_j \in S_1\} \cap \{j \mid v_j \in S_2\}$ since both $\{j \mid u_j \in S_1\}$ and $\{j \mid v_j \in S_2\}$ contain more than half of the elements in $\{1, \dots, 2k + 1\}$. Thus $(u_i, v_i) \in S_1 \times S_2$, and the order of $\mathcal{I}_{12}$ is at most $2k + 1$.
- (ii) Let $\mathcal{J}_i$ be the family $\{\{w\} \mid w \in V(G_i)\}$ for $i \in \{1, 2\}$. Thus $\mathcal{J}_{12}$ is the set $\{(u, v) \mid u \in V(G_1), v \in V(G_2)\}$. Clearly, two sets $\{(u_i, v_i)\}, \{(u_j, v_j)\} \in \mathcal{J}_{12}$ are not touched in $G_1 \square G_2$ if $i \neq j$. Hence $\mathcal{J}_{12}$ is not a bramble of $G_1 \square G_2$. $\square$

4. Applications

In this section, we present some applications of our lower bounds. The bounds are best possible in the sense that they are tight for some cases.

4.1. Bounds for strong product

As mentioned earlier, Bodlaender, Gilbert, Hafsteinsson, and Kloks [4, Theorem 21] implicitly showed that $\text{bn}(G \boxtimes K_n) = \text{bn}(G) \cdot n$ for any graph G . As $\eta(K_n) = |V(K_n)| = n$, our lower bound $\text{bn}(G_1 \boxtimes G_2) \geq \text{bn}(G_1) \cdot \eta(G_2)$ can be regarded as a generalization of this fact with the upper bound $\text{bn}(G_1 \boxtimes G_2) \leq \text{bn}(G_1) \cdot |V(G_2)|$. To show this upper bound, let $(\{X_i \mid i \in V(T)\}, T)$ be a tree-decomposition of G_1 with $\max_i |X_i| = \text{bn}(G_1)$. Then $(\{X_i \times V(G_2) \mid i \in V(T)\}, T)$ is a tree-decomposition of $G_1 \boxtimes G_2$. The first and third conditions are clearly satisfied. Let $\{(u_1, v_1), (u_2, v_2)\}$ be an edge of $G_1 \boxtimes G_2$. Because $u_1 = u_2$ or $\{u_1, u_2\} \in E(G_1)$, there is some i such that $\{u_1, u_2\} \subseteq X_i$. Therefore, $\{u_1, u_2\} \times \{v_1, v_2\} = \{(u_1, v_1), (u_2, v_2)\} \subseteq X_i \times V(G_2)$. As $\max_i |X_i \times V(G_2)| = \text{bn}(G_1) \cdot |V(G_2)|$, the bound holds.

4.2. Cartesian product of two complete graphs

As mentioned in Section 1.3, Lucena [18] determined the treewidth of the Cartesian product of two complete graphs of n vertices. More precisely, it was shown that $\text{bn}(K_n \square K_n) = n^2/2 + n/2$ for any n . Since $\iota(K_n) = \lceil n/2 \rceil$ and $\eta(K_n) = n$, our lower bound is

$$\iota(K_n) \cdot \eta(K_n) = \lceil n/2 \rceil \cdot n = \begin{cases} n^2/2 & \text{if } n \text{ is even,} \\ n^2/2 + n/2 & \text{if } n \text{ is odd.} \end{cases}$$

Thus, for odd n , our bound is tight and can be considered as a generalization of the lower bound on $\text{bn}(K_n \square K_n)$. Note that for even n , our lower bound $n^2/2$ is not tight. Lucena constructed a specialized bramble of order $n^2/2 + n/2$ for even n .

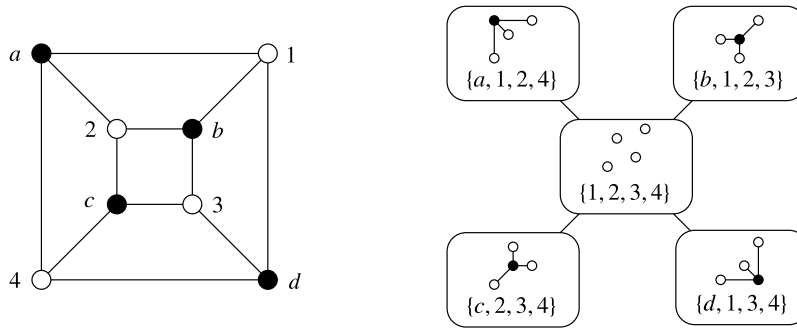
4.3. Cartesian product of a complete graph and a grid

Using Theorem 3.1, we obtain Theorem 4.1 given below.

Theorem 4.1. Let $\Gamma_{m,n}$ be the $m \times n$ grid with $m \geq n$. Then $\text{bn}(K_\ell \square \Gamma_{m,n}) \in \{\ell n, \ell n + 1\}$.

Proof. First we show that $\text{bn}(K_\ell \square \Gamma_{m,n}) \leq \ell n + 1$. It is known that $\text{bn}(P_n \square G) \leq |V(G)| + 1$ for any graph G (see e.g. [10]). As $K_\ell \square \Gamma_{m,n}$ is isomorphic to $(K_\ell \square P_n) \square P_m$, it follows that $\text{bn}(K_\ell \square \Gamma_{m,n}) = \text{bn}((K_\ell \square P_n) \square P_m) \leq |V(K_\ell \square P_n)| + 1 = \ell n + 1$.

Next, we show that $\text{bn}(K_\ell \square \Gamma_{m,n}) \geq \ell n$. Clearly, $\eta(K_\ell) = \ell$. We now show that $\iota(\Gamma_{m,n}) \geq n$ when $m \geq n$. Let $C_{i,j} = \{(i', j') \in V(\Gamma_{m,n}) \mid i' = i \text{ or } j' = j\}$ and $\mathcal{C} = \{C_{i,j} \mid 1 \leq i, j \leq n\}$. It is evident that $\mathcal{C}$ is an intersecting family. Suppose that there is a hitting set H of $\mathcal{C}$ with $|H| < n$. By the pigeon-hole principle, there exist indices i and j such that H does not intersect with the i th row and the j th column of $\Gamma_{m,n}$. Thus H cannot intersect with $C_{i,j}$. This contradicts that H is a hitting set of $\mathcal{C}$. By Theorem 3.1, it follows that $\text{bn}(K_\ell \square \Gamma_{m,n}) \geq \ell n$. $\square$

Fig. 4. A tree-decomposition of $P_2 \square P_2 \square P_2$.

The lower and upper bounds in the theorem above have tight examples. By applying Theorem 4.1, we have $\text{bn}(K_2 \square \Gamma_{2,2}) \in \{4, 5\}$, and clearly $\text{bn}(K_2 \square \Gamma_{2,2}) = \text{bn}(P_2 \square P_2 \square P_2) \leq 4$ (see Fig. 4). By Theorem 4.1, $\text{bn}(K_2 \square \Gamma_{2n,n}) \in \{2n, 2n + 1\}$. Since $K_2 \square \Gamma_{2n,n}$ is isomorphic to $P_2 \square P_n \square P_{2n}$, and $P_2 \square P_n$ contains P_{2n} as a subgraph, $K_2 \square \Gamma_{2n,n}$ contains $\Gamma_{2n,2n}$ as a subgraph. Therefore, $\text{bn}(K_2 \square \Gamma_{2n,n}) \geq \text{bn}(\Gamma_{2n,2n}) = 2n + 1$ (see [3]).

4.4. Cartesian product of a complete graph and a complete multipartite graph

Next, we present lower and upper bounds for the treewidth of the Cartesian product of a complete graph and a complete multipartite graph.

Theorem 4.2. If $n_1 = \max\{n_i \mid 1 \leq i \leq k\}$ and $n = \sum_{i=1}^k n_i$, then

$$\ell \cdot (n - \max\{n_1, \lfloor n/2 \rfloor\}) \leq \text{bn}(K_\ell \square K_{n_1, \dots, n_k}) \leq \ell \cdot (n - n_1 + 1).$$

Proof. Djelloul [10, Fact 3.1(i)] showed that $\text{bn}(G \square H) \leq \min\{\text{bn}(G) \cdot |V(H)|, \text{bn}(H) \cdot |V(G)|\}$ for any G and H . The upper bound $\ell \cdot (n - n_1 + 1)$ follows from this result and the fact $\text{bn}(K_{n_1, \dots, n_k}) = n - n_1 + 1$ [8].

Now, we present the lower bound. Let $n' = \max\{n_1, \lfloor n/2 \rfloor\}$. Since $\eta(K_\ell) = \ell$, it suffices to show that $\iota(K_{n_1, \dots, n_k}) \geq n - n'$ by Theorem 3.1. Note that any subset of $V(K_{n_1, \dots, n_k})$ with more than n' vertices is connected and contains more than half of the vertices of $K_{n_1, \dots, n_k}$. Hence, for any $X \subset V(K_{n_1, \dots, n_k})$ with $|X| < n - n'$, it holds that $V(K_{n_1, \dots, n_k}) - X$ is connected and $|V(K_{n_1, \dots, n_k}) - X| > |V(K_{n_1, \dots, n_k})|/2$. Therefore, $V(K_{n_1, \dots, n_k})$ is an $(n - n')$ -linked set. This implies that $\iota(K_{n_1, \dots, n_k}) \geq \lambda(K_{n_1, \dots, n_k}) \geq n - n'$. $\square$

5. Conclusion

We have presented two lower bounds for the treewidth of product graphs. This forms, in a way, the first step in the study of the inapproximability of the bramble number (and the treewidth) of graphs. To present some applications of these results, we determined the treewidth of some product graphs. The lower bounds are tight in some cases.

Some problems remain open. The (in)approximability of treewidth is still unsettled. Since resolving the Unique Games Conjecture or the Small Set Expansion conjecture would be very hard, we believe that our direction would be useful. (Recall that Austin, Pitassi, and Wu [2] showed that treewidth is hard to approximate within any constant factor assuming the Small Set Expansion conjecture.) Also, the complexity of the problem for determining the PI number of a graph remains open. We think that this problem would be NP-hard.

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